

```
In [6]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [7]: df = pd.read_csv('Expanded_data_with_more_features.csv.zip')
```

```
In [8]: df.head()
```

```
Out[8]:
```

	Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMar
0	0	female	NaN	bachelor's degree	standard	none	
1	1	female	group C	some college	standard	NaN	
2	2	female	group B	master's degree	standard	none	
3	3	male	group A	associate's degree	free/reduced	none	
4	4	male	group C	some college	standard	none	

```
In [9]: df.describe()
```

```
Out[9]:
```

	Unnamed: 0	NrSiblings	MathScore	ReadingScore	WritingScore
count	30641.000000	29069.000000	30641.000000	30641.000000	30641.000000
mean	499.556607	2.145894	66.558402	69.377533	68.418622
std	288.747894	1.458242	15.361616	14.758952	15.443525
min	0.000000	0.000000	0.000000	10.000000	4.000000
25%	249.000000	1.000000	56.000000	59.000000	58.000000
50%	500.000000	2.000000	67.000000	70.000000	69.000000
75%	750.000000	3.000000	78.000000	80.000000	79.000000
max	999.000000	7.000000	100.000000	100.000000	100.000000

```
In [10]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30641 entries, 0 to 30640
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Unnamed: 0            30641 non-null  int64  
 1   Gender                30641 non-null  object  
 2   EthnicGroup           28801 non-null  object  
 3   ParentEduc            28796 non-null  object  
 4   LunchType             30641 non-null  object  
 5   TestPrep              28811 non-null  object  
 6   ParentMaritalStatus   29451 non-null  object  
 7   PracticeSport         30010 non-null  object  
 8   IsFirstChild          29737 non-null  object  
 9   NrSiblings            29069 non-null  float64 
10   TransportMeans        27507 non-null  object  
11   WklyStudyHours        29686 non-null  object  
12   MathScore             30641 non-null  int64  
13   ReadingScore          30641 non-null  int64  
14   WritingScore          30641 non-null  int64  
dtypes: float64(1), int64(4), object(10)
memory usage: 3.5+ MB

```

```
In [12]: df.isnull().sum()
```

```

Out[12]: Unnamed: 0            0
         Gender              0
         EthnicGroup        1840
         ParentEduc        1845
         LunchType          0
         TestPrep          1830
         ParentMaritalStatus 1190
         PracticeSport       631
         IsFirstChild        904
         NrSiblings         1572
         TransportMeans      3134
         WklyStudyHours      955
         MathScore           0
         ReadingScore        0
         WritingScore        0
         dtype: int64

```

```
In [14]: df = df.drop('Unnamed: 0',axis =1)
```

```
In [17]: df.sample(2)
```

```

Out[17]:

```

	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus
2116	female	group E	master's degree	standard	none	divorced
6462	female	group C	master's degree	standard	none	married

```
In [18]: df['WklyStudyHours'] = df['WklyStudyHours'].str.replace('5 - 10','5 - 9')
```

```
In [19]: df['WklyStudyHours'].head()
```

```
Out[19]: 0      < 5
          1      5 - 9
          2      < 5
          3      5 - 9
          4      5 - 9
          Name: WklyStudyHours, dtype: object
```

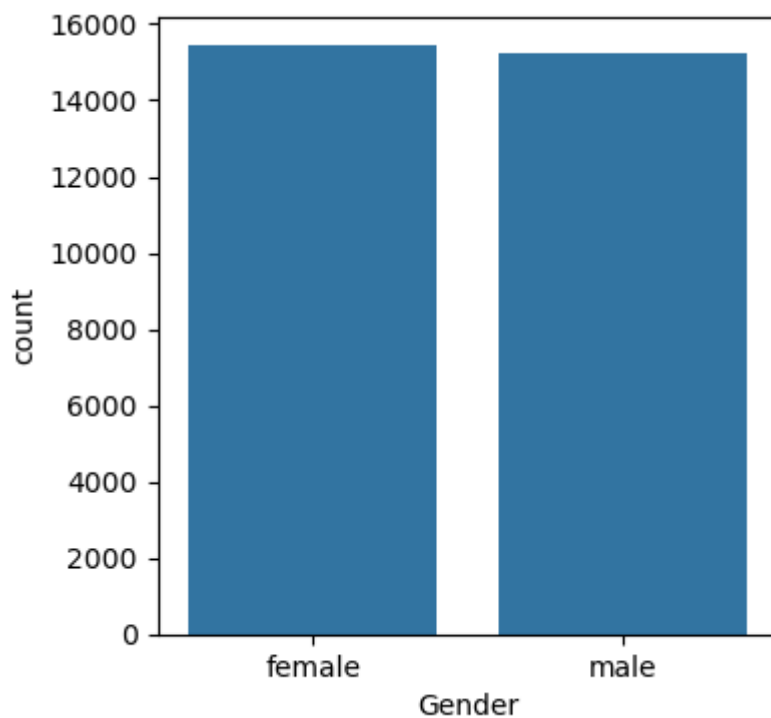
GENDER DISTRIBUTION

```
In [21]: df['Gender'].head()
```

```
Out[21]: 0      female
          1      female
          2      female
          3      male
          4      male
          Name: Gender, dtype: object
```

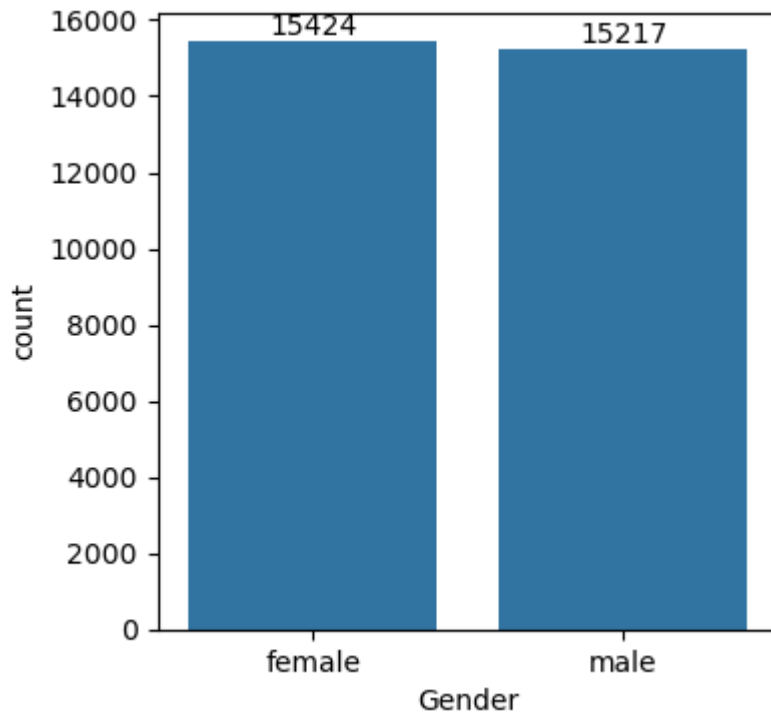
```
In [25]: #plotting count plot
plt.figure(figsize=(4,4))
sns.countplot(data = df, x = 'Gender')
plt.show
```

```
Out[25]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [26]: # plotting bar_label
plt.figure(figsize=(4,4))
ax = sns.countplot(data = df, x = 'Gender')
ax.bar_label(ax.containers[0])
plt.show
```

```
Out[26]: <function matplotlib.pyplot.show(close=None, block=None)>
```



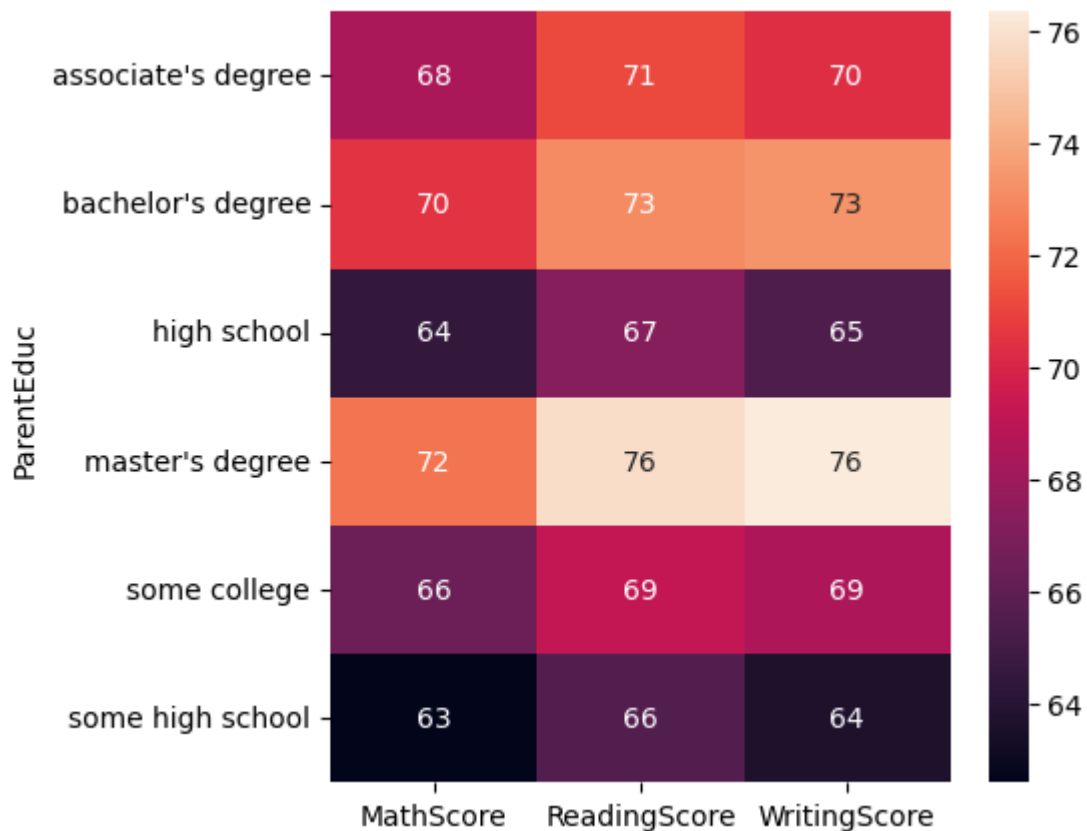
```
In [ ]: # from the above graph we have analysed that the count of female is more
        female = 15424
        male = 15217
```

ANALYSISING STUDENT MARKS BASED ON PARENTS EDUCATION

```
In [27]: gb = df.groupby('ParentEduc').agg({"MathScore": "mean", "ReadingScore": "mea
```

```
In [32]: #plotting heatmap
        plt.figure(figsize=(5,5))
        sns.heatmap(gb, annot = True)
        plt.show
```

```
Out[32]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [ ]: # from the above graph we can conclude that the parents who have complete  
# thier students are scoring very good marks
```

ANALYSING STUDENT MARKS BASED ON ParentMaritalStatus

```
In [35]: gb1 = df.groupby('ParentMaritalStatus').agg({"MathScore": "mean", "ReadingS
```

```
In [36]: #plotting heatmap  
plt.figure(figsize=(5,5))  
sns.heatmap(gb1, annot = True)  
plt.show
```

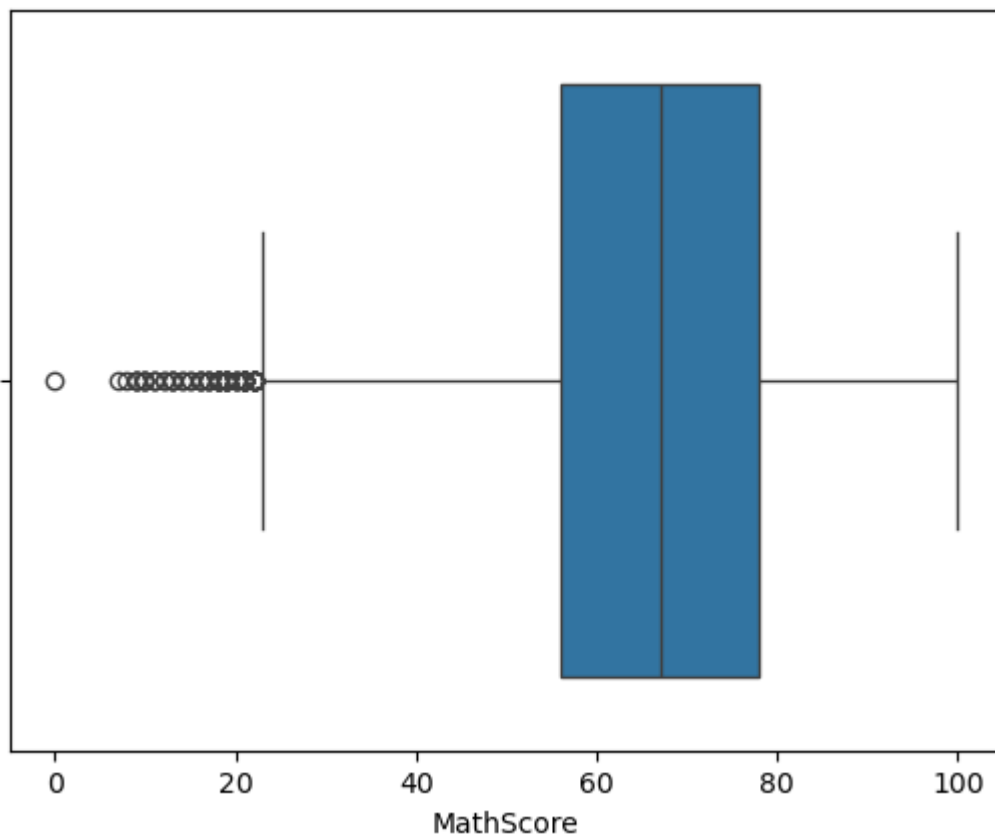
```
Out[36]: <function matplotlib.pyplot.show(close=None, block=None)>
```



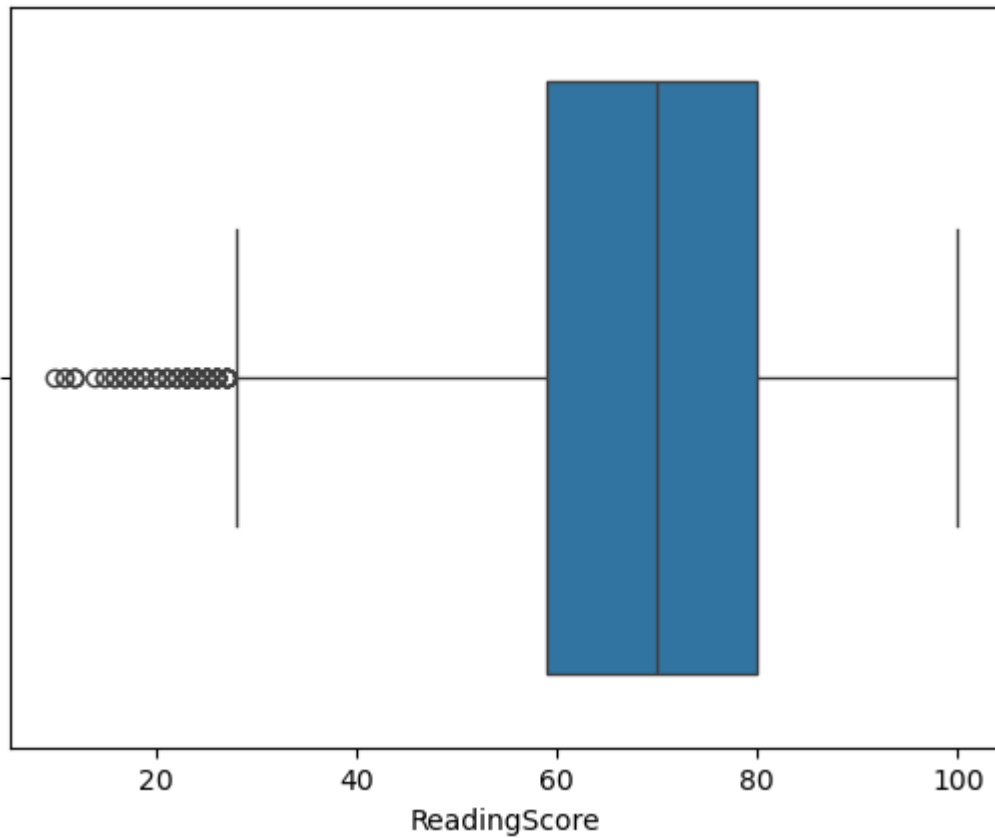
```
In [ ]: # FROM THE ABOVE GRAPH WE HAVE ANALYSED THAT THERE IS NO/NEGLIGBLE MARKS
```

box plot for checking outlier

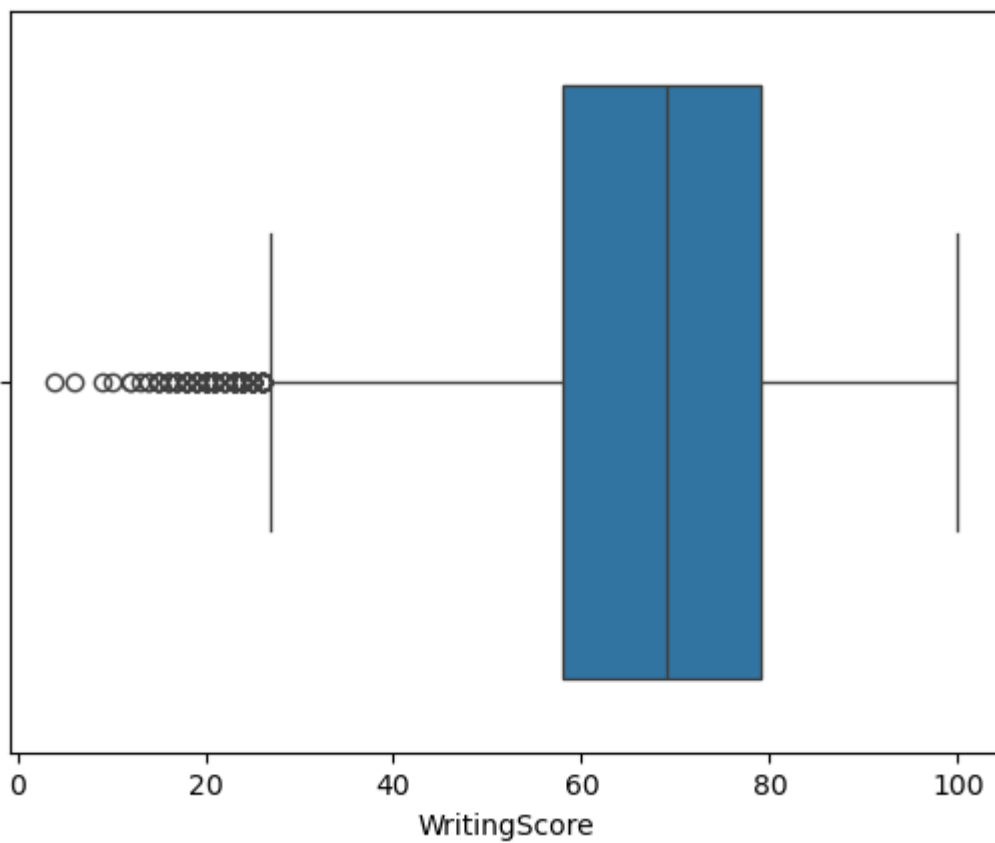
```
In [39]: sns.boxplot(data = df, x = "MathScore")
plt.show()
```



```
In [41]: sns.boxplot(data = df,x ="ReadingScore")  
plt.show()
```



```
In [42]: sns.boxplot(data = df,x ="WritingScore")  
plt.show()
```



```
In [ ]: #there are outliers present in the data
```

